EIGEN · CHARACTERISTIC POLYNOMIAL

What the polynomial encodes

The roots give the eigenvalues, which is the reason to form it. The coefficients give trace and determinant for free, which is the reason it is worth reading rather than only solving.

9
READINGS

WHAT THE COEFFICIENTS CARRY 3

1 **Degree and leading term**

the sign alternates with n

$\deg p = n$, leading term $(-1)^n \lambda^n$

One root per dimension, counted with multiplicity, over $\mathbb{C}$. Some texts negate the whole polynomial to make it monic — harmless, but it flips the sign of every coefficient below, so the convention has to be fixed before the formulas are quoted.

2 **Coefficient of λ^{n-1}**

with algebraic multiplicity

$$\sum \lambda_i = \operatorname{tr}(A)$$

The trace is available without solving anything — it is the sum of the diagonal, and the polynomial says that sum is also the sum of the roots. A cheap consistency check on any eigenvalue computation.

3 **Constant term**

always

$$p(0) = \det(A) = \prod \lambda_i$$

Setting $\lambda = 0$ in $\det(A - \lambda I)$ leaves $\det(A)$. So A is invertible exactly when $p(0) \neq 0$, which is the same as saying no eigenvalue is zero — three statements that turn out to be one.

WHAT THE ROOTS CARRY 3

4 **Roots are the eigenvalues**

over $\mathbb{C}$, with multiplicity

$p(\lambda) = 0 \iff \lambda$ is an eigenvalue

The determinant condition is what turns "$(A - \lambda I)\mathbf{x} = \mathbf{0}$ has a nontrivial solution" into an equation that can be solved. Over $\mathbb{R}$ some roots may not exist — a rotation has none — which is why the spectrum is stated over $\mathbb{C}$.

5 **Algebraic multiplicity**

$m_g \leq m_a$, never the reverse

multiplicity of λ as a root

How many times a root repeats, which is not the same as how many independent eigenvectors it has. When the geometric count falls short the matrix is defective and cannot be diagonalized — the polynomial cannot detect this on its own.

6 **Similarity invariance**

P invertible

$$p_{P^{-1}AP}(\lambda) = p_A(\lambda)$$

Similar matrices share the polynomial, hence the eigenvalues, trace and determinant. Two matrices with different characteristic polynomials cannot represent the same transformation in different bases — which makes p a usable test for similarity in one direction.

CLOSED FORMS AND CONSEQUENCES 3

7 **2×2 form**

worth memorising

$$p(\lambda) = \lambda^2 - \operatorname{tr}(A)\lambda + \det(A)$$

The discriminant $\operatorname{tr}^2 - 4\det$ classifies the pair before any solving: positive gives two distinct reals, zero gives a repeated root, negative gives a complex conjugate pair.

8 **3×3 form**

M_2 = sum of principal 2×2 minors

$$-\lambda^3 + \operatorname{tr}(A)\lambda^2 - M_2\lambda + \det(A)$$

The pattern continues: each coefficient is a sum of principal minors of one size. Beyond 3×3 the expansion is not worth doing by hand, and numerical work does not compute the polynomial at all.

9 **Cayley–Hamilton**

every square matrix satisfies its own polynomial

$$p(A) = O$$

Substituting the matrix into its own characteristic polynomial gives the zero matrix. The practical consequence is that A^n can be rewritten in terms of lower powers, so any power or the inverse is a polynomial in A of degree below n.

Every entry is a property of the transformation rather than of the matrix expressing it: similar matrices have the same polynomial, so all of trace, determinant and spectrum survive a change of basis. That invariance is what makes the polynomial a description of the map and not merely of the array.